

Fursah: Explainable AI for Workforce Readiness

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Executive decision

Fursah is a working education-to-employment decision-support prototype built around a strict boundary: deterministic, versioned rules produce every consequential score; generative AI proposes document fields and explains authorized platform facts; humans verify evidence and retain every consequential decision. The contribution is both a use case and a reviewable AI-readiness package: Y.3172 traceability, a 13-dimension self-assessment, 17 authentic knowledge-base entries, and six operational policy recommendations.

7/7	13/13	17
Y.3172 nodes mapped	dimensions assessed	authentic sources

The problem and intervention

Students receive opaque rejections, employers screen through weak proxies, and universities learn about skill mismatch only after curriculum cycles have passed. Fursah turns employer requirements, evidence-backed student profiles and university offerings into a shared skill language. It produces readiness explanations and next actions for students, job-related fit explanations for employers, and privacy-suppressed demand signals for universities.

Implemented proof, not claimed impact

The prototype includes prepared synthetic accounts. It does not claim improved placement rates, reduced time-to-hire or independent fairness validation. Those are post-deployment outcomes requiring real participants and governed evaluation. Production remains blocked until hosting, processor and cross-border-transfer controls are institutionally approved.

Criterion 1 - Clear Y.3172 use-case pipeline

The node labels and functions below follow ITU-T Y.3172 clause 8.1. Fursah implementation text is deliberately kept separate from the standard function. Source traceability is published at fursah.org/standards.

Node	Standard role	Fursah implementation	Governing instruments
SRC	Source	Evidence, role requirements, university offerings	PDPL; NDMO classification
C	Collector	Role-scoped APIs, sessions and institutional ingestion	NCA ECC/CCC; DGA interoperability
PP	Preprocessor	Document extraction, taxonomy normalization, consent	PDPL minimization; DPIA
M	Model	Deterministic scoring plus grounded language model	SDAIA AI Ethics; ISO 42001/23894
P	Policy	Consent, review thresholds, suppression and override	PDPL rights; human oversight
D	Distributor	Role-authorized output; cohorts suppressed below five	National Data Governance
SINK	Sink	Student, employer, university and policy interfaces	WCAG 2.1 AA; Arabic-first

The critical boundary at M

Readiness, gap and candidate-role scores are calculated from published weights. The production assistant was verified on 21 August 2026 returning a grounded explanation through Cloudflare Workers AI using Llama 3.1 8B; it repeated the same component figures displayed by the dashboard and disclosed its grounding versions. The assistant cannot calculate or alter a score, ranking or evidence decision.

Criterion 2 - Mapping to all 13 AI Readiness dimensions

Coverage is intentionally not universal. Addressed means implemented in the prototype; Partial means an enabling component exists but the full dimension exceeds application scope; Out of scope means Fursah makes no claim.

#	Dimension	Coverage	Fursah evidence or limitation
1	Data/model Marketplace	Partial	Shared, versioned skill taxonomy; no marketplace or monetization layer.
2	Generated Content Marketplace	Out of scope	No tradeable generated dataset or model asset is produced.
3	Cross-domain correlation analysis	Partial	Connects higher-education supply with employer demand.
4	Contextualization and Regional Impact	Addressed	Saudi taxonomy, Arabic interface, PDPL/SDAIA/NDMO/NCA controls.
5	Level of Integration of AI in Workflows	Addressed	Extraction, readiness, matching and curriculum alignment have named inputs, outputs and downstream humans.
6	Human Interface	Addressed	Bilingual runtime, accessibility target and role-scoped conversational explanations.
7	Strategy Alignment	Addressed	Students, employers and universities coordinate through one intelligence layer aligned to HCDP.
8	Collaboration with AI	Addressed	Humans approve extraction, may reject direction suggestions, and decide appeals and overrides.
9	Impacts of Humans in AI Integration	Addressed	Skills-gap analysis at student, institution and ecosystem resolutions.
10	AI and Policies	Addressed	Governance scenarios, audit events and policy-gap evidence support decision makers.
11	AI for Inclusion	Addressed	No gender, nationality, age or GPA enters scoring; published job-related criteria apply consistently.
12	Granular Priorities	Partial	Per-track and institution priorities exist; regional/sector weighting is future work.
13	Digital Infrastructure	Partial	Pipeline nodes are mapped; compliant production hosting remains a declared blocking gap.

Dimension 9 is the central contribution

The AI Ready Report explicitly identifies skills-gap analysis as a desired framework output. Fursah computes gaps at three resolutions: a student against a target role, an institution against employer demand, and the ecosystem as requested skills not covered by university offerings.

Criterion 3 - Contribution to the AI-RE knowledge base

The inclusion rule is strict: every entry must be publicly checkable and load-bearing in the solution. The live register identifies publisher, edition, language, official URL, what the source contains, what in Fursah depends on it, and the source files it constrains. Structured JSON and CSV exports are available from fursah.org/api/knowledge-base.

Representative source	Contribution	Official location
ITU-T Y.3172 (06/2019)	Pipeline vocabulary and clause 8.1 node mapping	itu.int/rec/T-REC-Y.3172-201906-I/en
ITU AI Ready Report 2.0 (2026)	13 dimensions and chapter 4 gap taxonomy	aiforgood.itu.int/event/ai-readiness-hackathon-kingdom-of-saudi-arabia/
Saudi Personal Data Protection Law	Minimization, rights, consent and transfer conditions	sdaia.gov.sa
SDAIA AI Ethics Principles	Fairness, explainability, accountability and human oversight	sdaia.gov.sa
NCA ECC and CCC	Cybersecurity and cloud-control baseline	nca.gov.sa/en/regulatory-documents/control-s-list/
DGA Digital Accessibility	WCAG 2.1 AA target and bilingual digital-service design	dga.gov.sa
Vision 2030 / HCDP	Education-to-labor-market alignment	vision2030.gov.sa
ISO/IEC 42001 and 23894	AI management and risk-management structure	iso.org

Review package fields

The JSON export additionally carries the complete Y.3172 pipeline, all 13 dimension assessments and all policy recommendations. This makes the contribution reusable and machine-reviewable rather than a list of links embedded only in prose.

Quality controls

The repository verification suite asserts seven unique pipeline identifiers, 13 ordered dimensions, explicit limitations, implementation evidence for every covered dimension, HTTPS source URLs, substantive source descriptions, and operational policy fields. Privacy, evidence and role-scoped assistant verification run separately.

Criterion 4 - Input to AI strategy and policy

Each recommendation below comes from a constraint encountered while building the prototype. It names an accountable public-sector owner, a trigger for application and a measurable outcome. Detailed rationale and review cadence are published at fursah.org/standards#gaps.

Encountered gap	Accountable owner	Operational trigger	Success measure
Fairness audit data	SDAIA/NDMO + MHRSD	Controlled pilot or 100 consequential recommendations	Separate audit attributes; quarterly proxy/disparity review
In-Kingdom inference	SDAIA + CST + approved cloud providers	Before identifiable production data reaches a model	All production inference/storage in approved region
National skill taxonomy	SDAIA + MoE + ETEC + skills councils	Cross-institution skill exchange	95% resolution to versioned national identifiers
Portable verified evidence	MoE + ETEC + credential issuers	Verified evidence leaves the reviewing system	Issuer, reviewer, method, date, status and revocation retained
Graduate outcome baseline	GASTAT + MoE + HCDP	Annual outcomes publication	Stable suppressed series with three comparable periods
Effective automation threshold	MHRSD + SDAIA	A score excludes or prioritizes a candidate	Human review evidence; override/appeal monitoring; suspension if absent

Policy position

Employment-adjacent AI should be governed by effective use, not product labels. If a ranking is used to exclude candidates without documented review, it functions as an automated decision even when called advisory. Fursah therefore recommends an operational threshold based on review evidence, override rates, appeals and suspension when controls are absent.

Evaluation scenarios and controls

Scenario	Pre-agreed response
Commercial capture	Sponsored employers or courses must be labeled, excluded from scores and auditable; suspend ranking if separation fails.
Inferred-attribute advertising	Prohibit inferred readiness, disability or socioeconomic signals from advertising; stop sharing and escalate under PDPL.
Potential disparate impact	Use separately governed audit data, investigate proxies and thresholds, document remediation, and suspend when necessary.
Automation bias	Require written review for exclusions, monitor unusually low override rates and retrain reviewers.
Non-standard candidate	Accept portfolios and practical evidence, describe low scores as evidence gaps, and provide appeal and override routes.
Small cohorts	Suppress groups below five, withhold statistics and prevent complementary disclosure through partitioning.

Verification evidence

The current suites pass privacy suppression, assistant context boundaries, evidence approval rules, reviewer attribution and submission completeness. Live behavioral assistant verification succeeded in the deployed prototype. Local behavioral probes remain environment-dependent because deployment secrets are intentionally not copied into the repository.

What remains external

Independent accessibility audit, governed pilot outcomes, expert validation of scoring weights, lawful fairness-audit data and approval of production hosting arrangements cannot be self-certified by the team. They are explicitly presented as validation work, not prototype achievements.

Judge demonstration - one proof chain

Step	Live evidence
1	Open fursah.org/judge-demo and select Abdullah Al-Ghamdi from prepared accounts.
2	Reconstruct 36/100 from the five weighted readiness components.
3	Ask why the score is what it is; confirm the response repeats displayed facts and declares its model/grounding versions.
4	Switch to Administrator; review AI extraction proposals, human verification and reviewer attribution.
5	Inspect appeals, overrides, cohort suppression and the operational Y.3172 trace in Governance.
6	Use employer and university accounts to follow skill demand into curriculum action.

System truth boundary

Generative AI may	Generative AI may not	Humans retain
Propose fields from evidence and explain authorized platform facts.	Calculate scores, rank people, verify evidence, change career direction, invent metrics or expose another role's data.	Evidence approval, hiring, curriculum decisions, appeals, overrides and production-risk acceptance.

Submission links

Prototype: <https://fursah.org> | Judge route: <https://fursah.org/judge-demo> | Standards: <https://fursah.org/standards> | Knowledge base: <https://fursah.org/knowledge-base> | Repository: <https://github.com/Shoug-Alomran/AI-Powered-Workforce-Readiness-Ecosystem>

Contact

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Closing statement

Fursah does not ask a judge to trust a black box. It asks the judge to inspect a pipeline, reconstruct a score, challenge an explanation, follow a human decision, and reuse the policy evidence behind it.